



Problem Statement #1

Disaster Prediction and Community Response System

Natural disasters such as floods, cyclones, landslides, and urban flooding can cause significant loss of life and property. Often, information about risk areas, shelters, emergency resources, and rescue requirements is fragmented across multiple sources.

Develop a **technology-driven disaster management platform** that can analyse available environmental, weather, geographical, and community-generated data to identify potential risks and support faster emergency response.

Expected Features

- Risk prediction or early-warning mechanism.
- Interactive map showing vulnerable locations.
- Identification of nearby shelters, hospitals, emergency services, and resources.
- Real-time notifications and alerts.
- Dashboard for administrators/rescue teams.
- Prioritization of rescue requests based on severity.
- Offline/low-connectivity functionality where possible.



Problem Statement #2

Healthcare Accessibility and Rural Health Assistant

People in rural and remote areas may face difficulties accessing timely healthcare information and appropriate medical facilities. Build a **non-diagnostic digital healthcare assistance platform** that helps users understand health-related information, locate appropriate healthcare resources, and manage basic healthcare activities.

The system should not replace doctors or provide definitive medical diagnoses.

Expected Features

- Multilingual conversational interface.
- Basic health-information guidance.
- Nearby hospital/health-center locator.
- Appointment/request management.
- Medicine and appointment reminders.
- Emergency contact functionality.
- Patient health-record organization.
- Voice-based interaction for users with limited literacy.
- Escalation to healthcare professionals when appropriate.



Problem Statement #3

Smart Agriculture: Crop Disease and Farm Advisory System

Farmers frequently face crop diseases, uncertain weather conditions, inefficient irrigation, and difficulty obtaining timely technical advice. Develop an intelligent platform that assists farmers in making data-driven decisions.

The system should combine available crop images, weather information, soil parameters, and agricultural knowledge to provide useful recommendations.

Expected Features

- Crop disease identification from images.
- Crop-health monitoring.
- Weather-based farming recommendations.
- Irrigation recommendations.
- Soil-health analysis.
- Pest/disease alerts.
- Regional-language interface.
- Farmer-friendly dashboard.
- Historical crop and farm analysis.



Problem Statement #4

AI Assistive Technology for Persons with Visual or Hearing Impairments

Many digital and physical environments remain difficult to navigate for people with disabilities. Develop an **AI-powered assistive system** that improves accessibility for visually or hearing-impaired users.

Teams may focus on one specific accessibility challenge rather than attempting to solve everything.

Expected Features

- Real-time object detection.
- Text/signboard recognition.
- Currency/document recognition.
- Text-to-speech.
- Speech-to-text.
- Environmental sound alerts.
- Navigation assistance.
- Obstacle detection.
- Accessible educational content.
- Emergency assistance.



Problem Statement #5

Cyber Shield: Intelligent Phishing, Scam and Cyber-Fraud Detection

Phishing links, fraudulent messages, fake websites, QR-code scams, and social-engineering attacks are increasingly targeting ordinary users. Develop an intelligent cybersecurity platform that can identify suspicious digital content and explain the associated risks.

Expected Features

The system may analyze:

- URLs.
- Emails.
- SMS messages.
- QR codes.
- Web pages.
- Social-media messages.

It should provide:

- Risk classification.
- Suspicious-pattern detection.
- Explanation of why content is risky.
- Safe browsing recommendations.
- Threat reporting.
- Educational awareness modules.